Tutorial 2: Linear Regression in R

Software: R *(Posit.cloud)*

By the end of the tutorial, you should be able to

- Specify a regression model from a research question.
- Estimate simple and multiple regressions with lm().
- Interpret coefficients, confidence intervals and model fit.
- Use heteroskedasticity-robust standard errors.
- Inspect residual and influence diagnostics.
- Distinguish association from causation in a policy briefing.

Files used

File	Purpose
country_context_tutorial.csv	Clean dataset from previous tutorial with one recent observation per country
country_context_codebook.xlsx	Definitions, units, providers, indicator codes and reference years
tutorial02_starter.R	Student script with section headings and incomplete commands

Part A. Open, inspect and verify

```
dat <- read.csv("data/country_context_tutorial.csv")
str(dat)
summary(dat)
colSums(is.na(dat))
sum(duplicated(dat$country_code))
```

- What does one row represent?
- Are all units and reference years comparable?
- Which variables contain missing observations?
- How many observations remain for a complete-case model?

Part B. Visualise the focal relationship

```
plot(youth_unemployment ~ log_gdp_pc_ppp, data = dat,
    xlab = "Log GDP per capita (PPP)",
    ylab = "Youth unemployment rate (%)")
abline(lm(youth_unemployment ~ log_gdp_pc_ppp, data = dat))
```

- Describe the direction, form and strength of the relationship.
- Identify countries that deserve investigation.
- Does this graph establish a causal relationship? Why not?

Part C. Estimate the first model

```
m1 <- lm(youth_unemployment ~ log_gdp_pc_ppp, data = dat)
summary(m1)
confint(m1)
```

- Interpret the slope in the variables' actual units.
- Interpret its 95% confidence interval.
- Is the intercept substantively meaningful?
- What does R-squared describe?

Part D. Add theoretically justified controls

```
m2 <- lm(youth_unemployment ~ log_gdp_pc_ppp +
         education_enrolment + internet_use +
         government_effectiveness + region, data = dat)
summary(m2)
confint(m2)
anova(m1, m2)
```

- Why was each control included?
- How did the focal coefficient change?
- What is the reference region?
- Does the comparison establish that Model 2 is causally correct?

Part E. Use robust uncertainty estimates

```
library(lmtest)
library(sandwich)
coeftest(m2, vcov = vcovHC(m2, type = "HC1"))
```

Compare the conventional and robust standard errors. The coefficients do not change; the estimated uncertainty may change.

- Which conclusions are sensitive to the standard-error choice?
- Why might residual variance differ across countries?

Part F. Inspect diagnostics

```
par(mfrow = c(2, 2))
plot(m2)
par(mfrow = c(1, 1))

influence.measures(m2)
sort(cooks.distance(m2), decreasing = TRUE)[1:5]
```

- Is there evidence of nonlinearity or changing variance?
- Which observations are influential?
- Should they be deleted automatically? Explain.
- What additional model or data check would you run?

Part G. Compare and communicate

Evidence	Model 1	Model 2
Focal coefficient		
Robust standard error		
95% confidence interval		
Observations		
R-squared		
Main limitation		

Policy paragraph

Imagine now you need to quickly brief a policy maker on your output. Write no more than 250 words. State the comparison, magnitude and uncertainty; identify the population and period; avoid causal language; name one data or model limitation; and identify one useful next analysis.